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Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

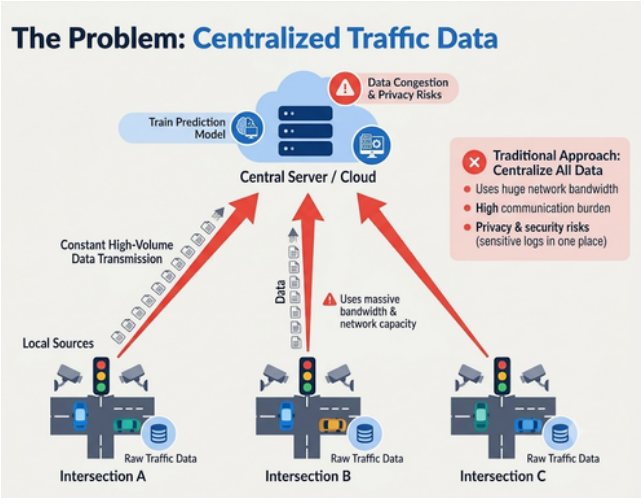
Traffic prediction systems rely on large amounts of data from cameras and roadside sensors.

TRADITIONAL APPROACH:

Sensors → Raw data → Central server → Prediction model

Continuously sending raw data to a central server:

- Consumes significant network bandwidth.
- Increases communication overhead.
- Concentrates potentially sensitive traffic and travel logs in one place.



Background

- Traffic prediction needs large amounts of data collected from different locations.
- Bringing all this data together can raise privacy and data sharing concerns.
- Federated learning allows different clients to train a model together without sharing their raw data.
- FedAvg combines what each client learns to create a shared global model.
- However, challenges such as different data distributions, communication cost, and model convergence still remain.
- Existing work, such as FedGRU, shows how can be applied to traffic prediction.
- **Our focus:** understand the trade off between prediction accuracy and communication cost, especially with IID and non IID data.

Optimization Focus:
Make federated traffic prediction more efficient by balancing accuracy, convergence, and communication cost.

Literature & RFC Review

FedAvg — McMahan et al. (2017)	Privacy-Preserving Traffic Flow Prediction — FedGRU (2020)	FedAGAT: Real-Time Federated Traffic Prediction	FedAGCN: Federated Traffic Prediction with Asynchronous Graph Convolution
• Allows multiple clients to train a shared model using their own local data.	• Applies federated learning to traffic flow prediction using a GRU based model.	• Uses graph attention to learn how traffic conditions at different locations influence one another.	• Combines federated learning with graph convolution to model road connections.
• The server combines locally trained model updates to create an improved global model.	• Learns traffic patterns from different locations without collecting their raw data centrally.	• Groups roads into communities and trains models locally before aggregating updates.	• Splits the traffic network into smaller subgraphs for local training.
• Reduces frequent communication with the server compared with traditional approaches.	• Demonstrates how federated learning can support collaborative traffic prediction.	• Evaluates prediction performance on two real-world traffic datasets and compares it with existing models.	• Uses asynchronous training to support clients updating at different times.

Proposed Solution

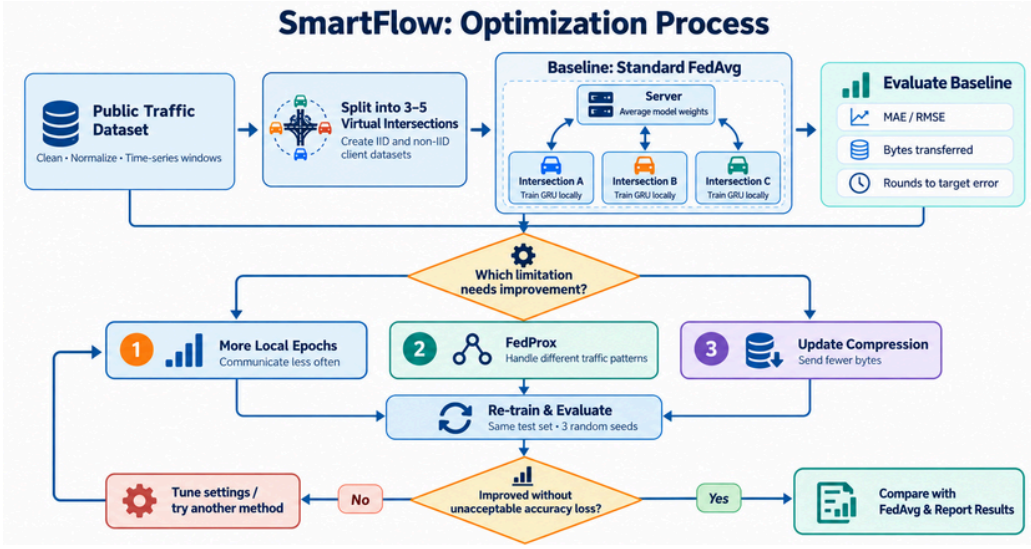
- **Federated Traffic Prediction**: Train a traffic-flow forecasting model locally at 3–5 simulated intersections instead of collecting all raw traffic data centrally.
- **FedAvg Aggregation**: Use Federated Averaging (FedAvg) to combine local model updates into a global prediction model.
- **Handle Data Heterogeneity**: Evaluate the model under IID and non-IID traffic distributions to reflect differences between intersections.
- **Communication-Aware Learning**: Reduce communication in FedAvg by tuning local training epochs (1, 3, or 5) while maintaining prediction accuracy.
- **Performance–Cost Trade-off**: Compare prediction accuracy (MAE/RMSE), convergence, and communication cost to determine whether communication can be reduced without significantly affecting prediction performance.



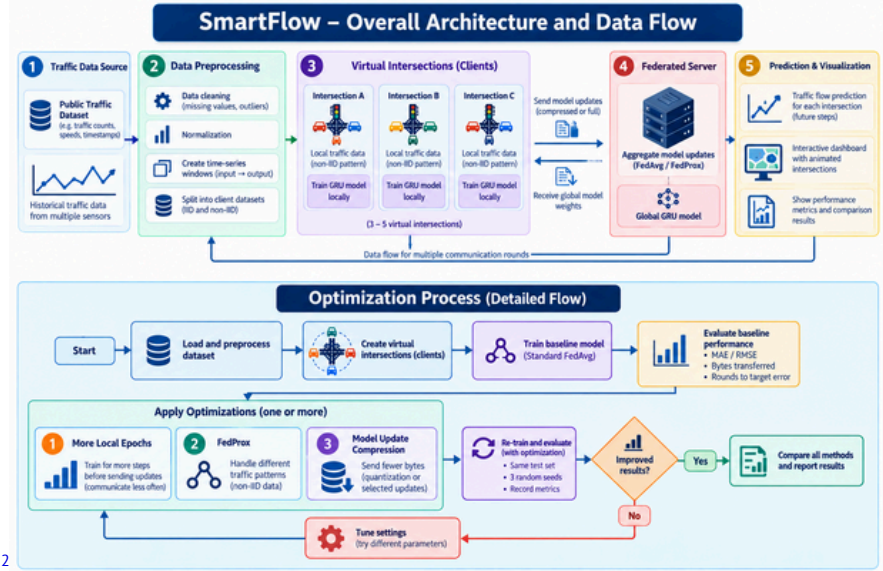
Optimization Proposed

Optimization	Before (Current Approach)	After (Proposed Improvement)
1. Communication efficient FedAvg	Each intersection trains its model locally and sends updates to the server after every communication round. Frequent communication can increase the total data exchanged between clients and the server.	Allow each intersection to train for more local epochs before sending its updates. This can reduce communication rounds and save network costs while maintaining good prediction accuracy.
2. Better learning with different traffic patterns (FedProx)	Different intersections may have different traffic volumes, rush hours, and congestion levels. Standard FedAvg may struggle to learn well when these local data patterns are very different (Non-IID data).	Use FedProx, which helps keep each client's model closer to the global model during local training. This may improve learning when intersections have different traffic patterns.
3. Model update compression	Each intersection sends full size model updates to the server, which can use a lot of communication bandwidth.	Compress model updates before sending them, Eg: by using fewer bits to represent values. This reduces the amount of data transferred.

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline



Bibliography I

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Thank You for Your Attention!